CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 15: Construction of Polygons

Topic-wise short notes with solved examples

Ruler-and-compass constructions of quadrilaterals, parallelograms, rectangles, rhombuses, squares, trapeziums and regular hexagons from given measurements.

1 Before you construct

- Draw a **rough sketch** first and mark the given measurements on it. It shows which triangle can be drawn first.
- A quadrilateral needs **five** independent measurements. Special quadrilaterals need fewer, because their properties supply the rest.

Figure	Measurements needed
Quadrilateral	5
Trapezium	4
Parallelogram	3
Rectangle	2
Rhombus	2
Square	1

- Use only a ruler and compasses unless a protractor is allowed. Leave all construction arcs visible.
- Standard angles by compasses: 60°, 120°, 90° (bisect a straight angle or use 60° and 120°), 30°, 45°, 75°, 105°, 135°.

2 Quadrilaterals

Given	Method
4 sides and 1 diagonal	Draw the triangle on the diagonal with two of the sides; draw the second triangle on the other side of the diagonal
3 sides and 2 diagonals	Draw the triangle formed by one side and parts fixed by the two diagonals from its ends; locate the fourth vertex by arcs
4 sides and 1 angle	Draw the two sides containing the angle; cut arcs for the other two sides from their free ends
3 sides and the 2 included angles	Draw the middle side; draw the angles at its ends; cut off the other two sides
2 adjacent sides and 3 angles	Draw the sides with their included angle; draw the other two angles at the free ends; the rays meet at the fourth vertex

Example 2.1 Construct quadrilateral ABCD with $AB = 4$ cm, $BC = 5$ cm, $CD = 4.5$ cm, $DA = 3.5$ cm and $AC = 6$ cm.

SOLUTION

Draw $AC = 6$ cm. With centres A and C and radii 4 cm and 5 cm, draw arcs meeting at B.

On the other side of AC, with centres A and C and radii 3.5 cm and 4.5 cm, draw arcs meeting at D.

Join AB, BC, CD and DA

Example 2.2 Can a quadrilateral be constructed with $AB = 3$ cm, $BC = 4$ cm and $AC = 8$ cm as one of its diagonals?

SOLUTION

In triangle ABC, $AB + BC = 7$ cm, which is less than AC.

No; triangle ABC cannot be drawn

3 Parallelograms

- Use: opposite sides are equal and parallel; the diagonals bisect each other.

Given	Method
2 adjacent sides and the included angle	Draw the sides with the angle; complete with arcs equal to the opposite sides
2 adjacent sides and 1 diagonal	Draw the triangle with the two sides and the diagonal; complete with arcs
2 diagonals and the angle between them	Draw one diagonal, mark its mid-point O; draw the other through O at the given angle, half on each side
1 side and 2 diagonals	Draw the triangle with the side and the two half-diagonals; produce each half-diagonal an equal length beyond O
2 adjacent sides and the height	Draw one side; draw a parallel at the given height; cut it with an arc equal to the other side

Example 3.1 Construct parallelogram ABCD with $AB = 6$ cm, $AC = 8$ cm and $BD = 6$ cm.

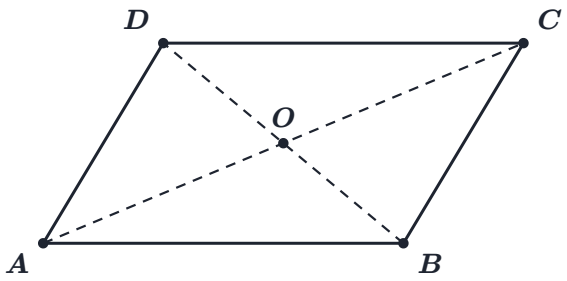
SOLUTION

The diagonals bisect each other at O, so $OA = 4$ cm and $OB = 3$ cm.

Draw $AB = 6$ cm. Draw triangle OAB with $OA = 4$ cm and $OB = 3$ cm.

Produce AO to C with $OC = 4$ cm and BO to D with $OD = 3$ cm.

Join BC, CD and DA



Draw triangle OAB first, then extend AO and BO to C and D

4 Rectangles

- Use: every angle is 90°; the diagonals are equal and bisect each other.

Given	Method
Length and breadth	Draw the length; draw perpendiculars at both ends; cut off the breadth
One side and a diagonal	Draw the side AB; draw a perpendicular at B; cut it from A with the diagonal as radius
Diagonal and the angle between the diagonals	Draw one diagonal, mark O; draw the other through O at the given angle, equal and bisected

Example 4.1 Construct rectangle ABCD with $AB = 5$ cm and diagonal $AC = 6.5$ cm.

SOLUTION

Draw $AB = 5$ cm. At B construct $\angle ABX = 90^\circ$.

With centre A and radius 6.5 cm, cut BX at C.

With centres A and C and radii BC and AB, draw arcs meeting at D; join AD and CD

5 Rhombuses and squares

- **Rhombus**: all sides equal; the diagonals bisect each other at 90°.

Given	Method
Side and one angle	Draw the two sides with the angle; complete with arcs equal to the side
Side and one diagonal	Draw the diagonal; draw arcs equal to the side from both ends, on both sides
Two diagonals	Draw one diagonal; draw its perpendicular bisector; mark half of the other diagonal on each side

- **Square**: one measurement is enough.

Given	Method
Side	Draw the side; draw perpendiculars at both ends; cut off the side
Diagonal	Draw the diagonal; draw its perpendicular bisector; mark half the diagonal on each side of the mid-point

Example 5.1 Construct a rhombus with diagonals 6 cm and 8 cm. What is its side?

SOLUTION

Draw $AC = 8$ cm and its perpendicular bisector, meeting AC at O. Mark $OB = OD = 3$ cm on the bisector.

Join AB, BC, CD, DA. Side$^2 = 4^2 + 3^2$.

The side measures 5 cm

Example 5.2 Construct a square with diagonal 5 cm.

SOLUTION

Draw $AC = 5$ cm and its perpendicular bisector through O.

Mark $OB = OD = 2.5$ cm on it and join AB, BC, CD, DA

6 Trapeziums

- A trapezium needs four measurements together with the fact that one pair of sides is parallel.
- **Method**: in trapezium ABCD with $AB \parallel DC$, draw CE parallel to DA to meet AB at E. Then AECD is a parallelogram, $AE = DC$, $CE = DA$ and $EB = AB - DC$. Construct triangle EBC first, then complete the parallelogram.

Example 6.1 Construct trapezium ABCD with $AB \parallel DC$, $AB = 7$ cm, $BC = 4$ cm, $CD = 4$ cm and $DA = 3.5$ cm.

SOLUTION

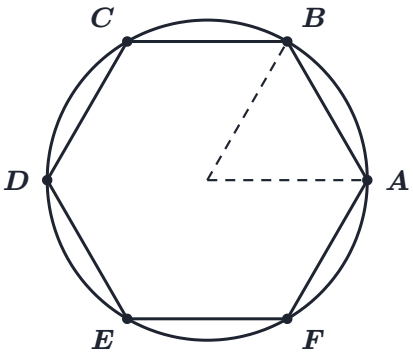
Draw $AB = 7$ cm and mark E on it with $AE = 4$ cm, so $EB = 3$ cm.

Draw triangle EBC with $EC = 3.5$ cm and $BC = 4$ cm.

With centres A and C and radii 3.5 cm and 4 cm, draw arcs meeting at D; join AD and CD

7 Regular hexagon

- Each interior angle of a regular hexagon is $120°$. Its side equals the radius of its circumscribed circle.
- **Method 1 (using the circle)**: draw a circle with radius equal to the side. Starting from any point on it, step off the radius six times round the circle. Join the points in order.
- **Method 2 (using angles)**: draw the side AB. At A and B construct angles of $120°$ and cut off the side on each arm. Repeat at the new vertices.



The side of a regular hexagon equals the radius of its circumcircle

Example 7.1 Construct a regular hexagon of side 3.5 cm.

SOLUTION

Draw a circle of radius 3.5 cm. Mark a point A on it.

With radius 3.5 cm, cut the circle at B from A, at C from B, and so on up to F.

Join AB, BC, CD, DE, EF and FA

8 Common mistakes

- Starting without a rough sketch and then finding that the first triangle cannot be drawn.
- Using the full diagonal instead of half when the diagonals bisect each other.
- Rubbing out the construction arcs.
- Measuring an angle with a protractor when a compass construction is required.
- Not checking the triangle inequality for the first triangle.
- Drawing the second triangle on the same side of the diagonal as the first.

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